CHARLOTTESVILLE HOUSING MARKET THROUGH THE YEARS

More listings preceded slower house-price growth

An exploratory study of mortgage rates, unemployment, and housing inventory

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Andrew Nguyen · Haero Lee · Hershi Aytoda

Group leader: Hershi Aytoda

OUTLINE

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Which factors track local house-price growth?

GOAL

Rank the factors associated with price growth and find the delay at which each one shows up.

RESEARCH QUESTION

Which factors are most strongly associated with the rise in Charlottesville house prices, and how quickly do prices respond to them?

Hypothesis

H_0: Holding the other predictors constant, changes in one predictor, have no association with year-over-year Charlottesville house price growth ($\beta = 0$).

H_1: Holding the other predictors constant, changes in one predictor, are associated with year-over-year Charlottesville house price growth ($\beta \neq 0$).

MODELING APPROACH

1. Convert every series to quarterly year-over-year changes.

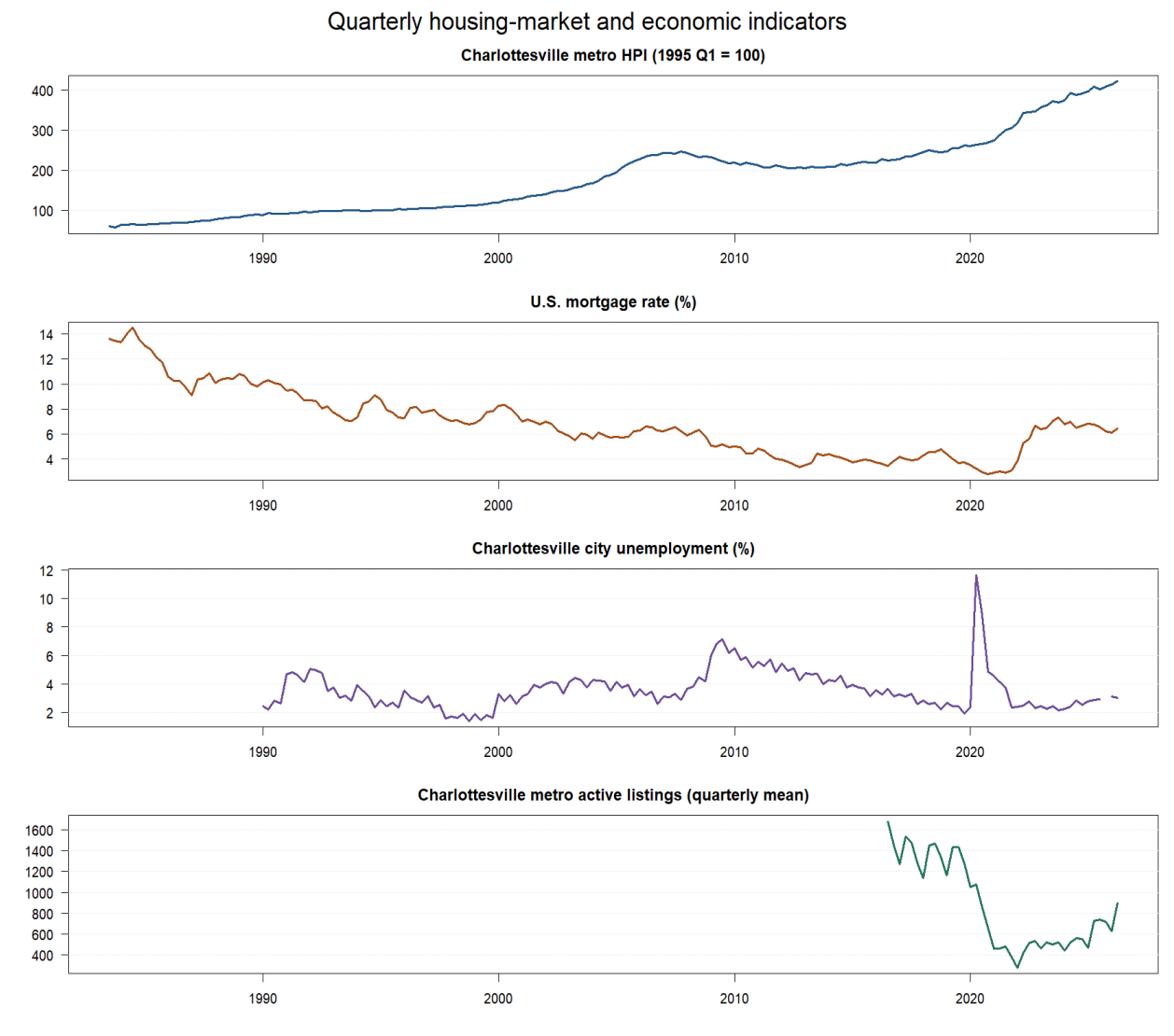
2. Select each predictor's delay, from 0 to 4 quarters.

3. Fit lagged multiple regression with Newey–West standard errors and standardized predictors.

4. Fit two models: long-run (mortgage, unemployment) and recent (adds listings).

H_0: a predictor's coefficient is zero. H_1: it is not (two-sided, $\alpha = 0.05$). -> Approach with a t-test and robustness checks

Which factors track local house-price growth?



Motivation and Context

1. Charlottesville is a small town, what makes it stand out is the University. University students and staff plus other locals create the housing market. How is this mixture reflected in the market?

2. Housing is the biggest purchase most people ever make, and in Charlottesville prices have climbed to r**oughly four times their 1995 level.** **Affordability** is usually discussed in terms of mortgage rates, but a few things move at once: borrowing costs, how strong the local job market is, and how many homes are actually for sale?

Four public FRED series, aligned by quarter

Series	Geography	Frequency	Annual change
House price index	Charlottesville metro	Quarterly	Percent
Mortgage rate	United States	Weekly	Percentage points
Unemployment	Charlottesville city	Monthly	Percentage points
Active listings	Charlottesville metro	Monthly	Percent

140

quarters in the long-run model (1991Q2–2026Q2)

31

quarters in the recent model (2018Q3–2026Q2)

FORMAT AND SIZE

Time Data, CSV files, merged into 172 quarterly rows (1983Q3–2026Q2). Listings begin in 2016.

ACQUISITION

Downloaded from FRED. With a Python script, averaged to quarters, no values imputed.

LICENSE AND ETHICS

Aggregate data, no individuals. Freddie Mac and Realtor.com terms apply: non-commercial use with citation. We report associations, not causes.

Series: ATNHPIUS16820Q, MORTGAGE30US, VACHAROURN, ACTLISCOU16820

Analysis pipeline and who built each part

1

Quarterly data

Andrew Nguyen

Download, year-over-year changes, preserve/adjust quarters

2

Exploratory analysis

Hershi Aytoda

Plots, coverage, distributions, visualize trends

3

Lagged regression

Haero Lee

Choose lags 0–4, fit two standardized models based on quarter

4

Evaluation

Hershi Aytoda

Diagnostics, robustness, results figures

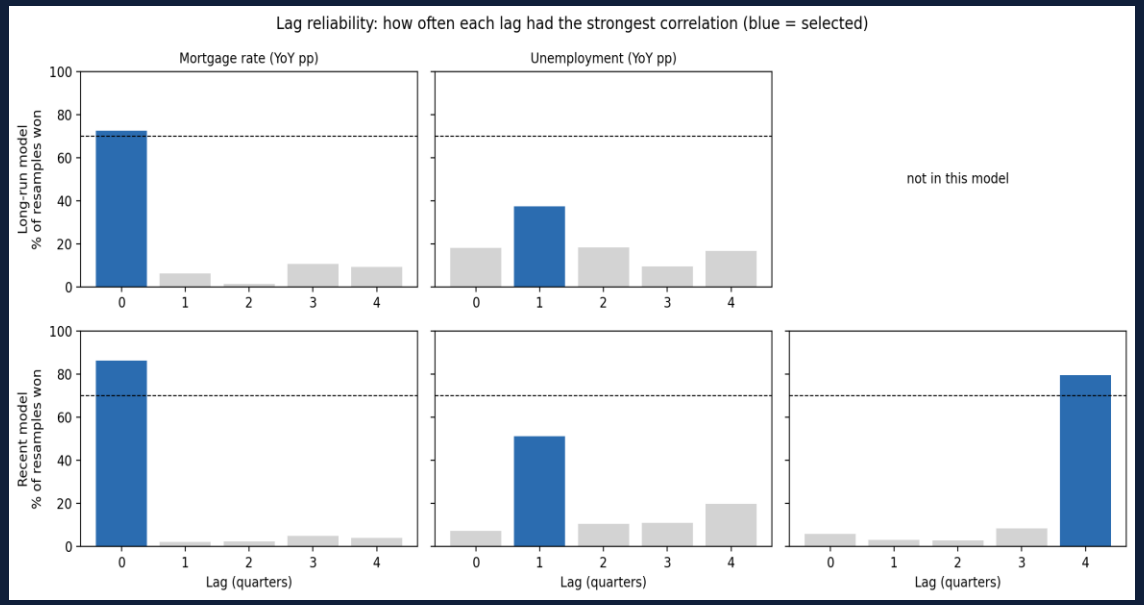
VALIDATION BUILT INTO THE PIPELINE

Residual diagnostics · alternative standard errors · bootstrap check of each lag · crisis-period and influence tests · robustness checks

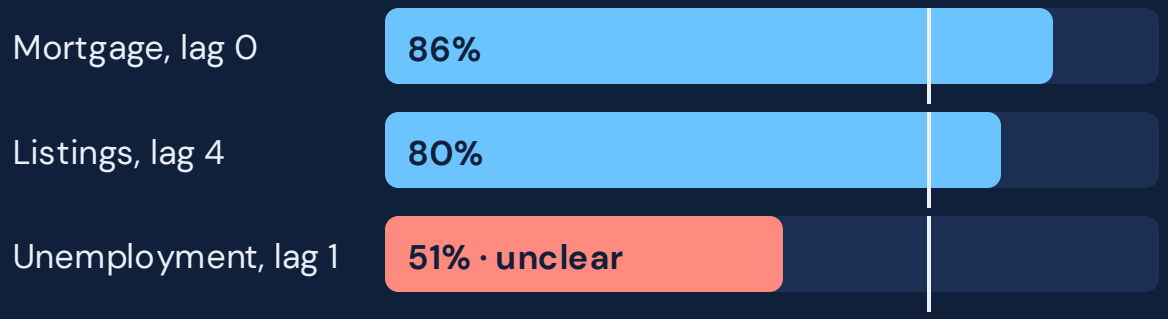
Key challenge: create calendar lags before removing incomplete model rows.

A tricky decision: how long is the delay?

We picked each lag by the strongest correlation across 0–4 quarters, then tested how stable that pick was.



How often each lag won across 2,000 resamples (recent model)



White marker = our 70% cutoff for a “clear” lag. Resamples use 8-quarter blocks.

WHY IT WAS HARD

Year-over-year changes overlap by three quarters, so neighboring lags look alike.

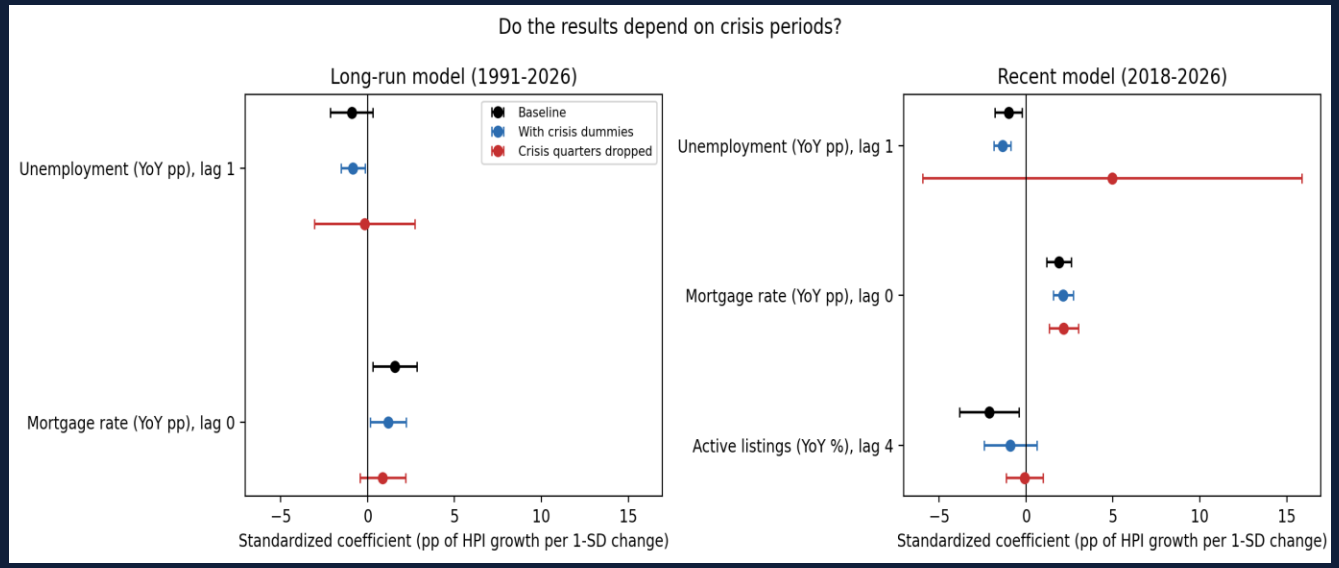
IMPACT

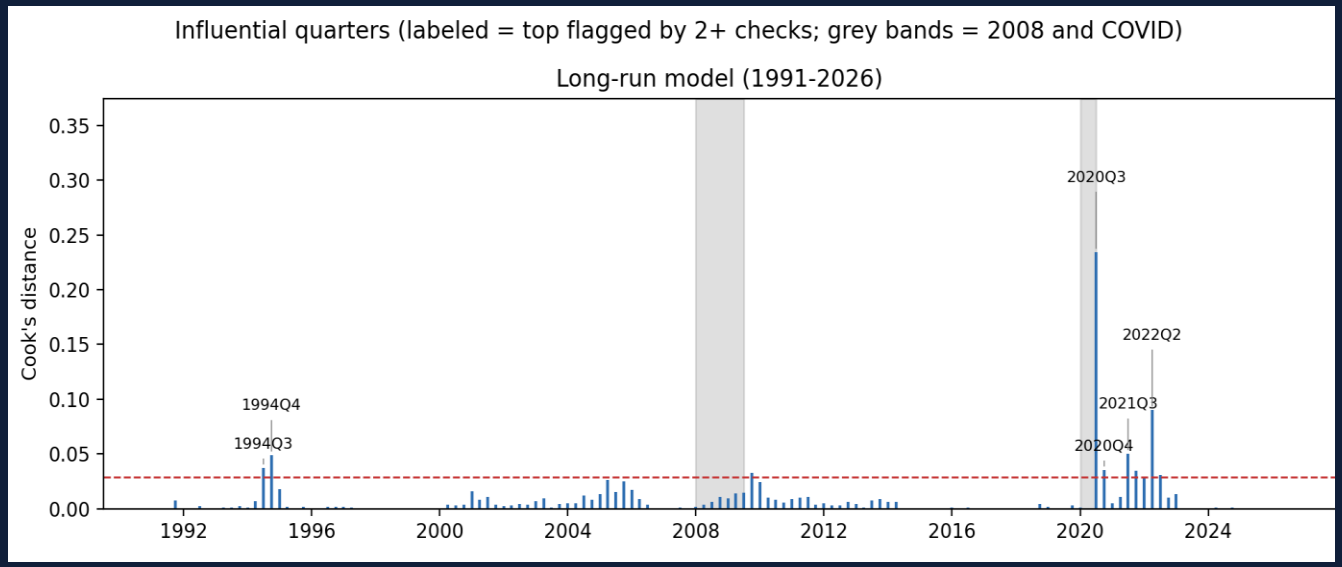
Unemployment timing is uncertain, so we report 0–2 quarters. Listings sits at the edge of our 0–4 range.

Biases and Uncertainties Considered

Concern	Test we ran	What it found
Long-run trend	Year-over-year changes	Removes most of the trend (a design step, not a test)
Unequal variance	Breusch–Pagan	p = 0.46 and 0.79: no problem
Overlapping predictors	VIF	1.04 and 1.32: no problem
Lags chosen on the same data	Bootstrap, 2,000 resamples	Mortgage (86%) and listings (80%) clear; unemployment (51%) not
Autocorrelation	Durbin–Watson, Ljung–Box	Durbin–Watson 0.20 and 0.66; Ljung–Box $p < 0.001$ and 0.002: errors stay correlated, so we used Newey–West intervals
Outlying quarters	Cook's distance	10 quarters flagged (long-run), 3 (recent), mostly 2020–23

How we measured uncertainty





95% INTERVALS

Newey–West errors. Significance conclusions hold with 6 or 8 lags and a small-sample correction.

CRISIS PERIODS

With crisis dummies, mortgage stays significant. Listings shrinks to −0.89 (p = 0.236).

WHAT STILL LIMITS US: Serial dependence, small sample, selected lags

Model fit summary

model	n_quarters	R2	adj_R2	n_significant_predictors
long_run	140	0.133	0.121	1
recent	31	0.692	0.658	3

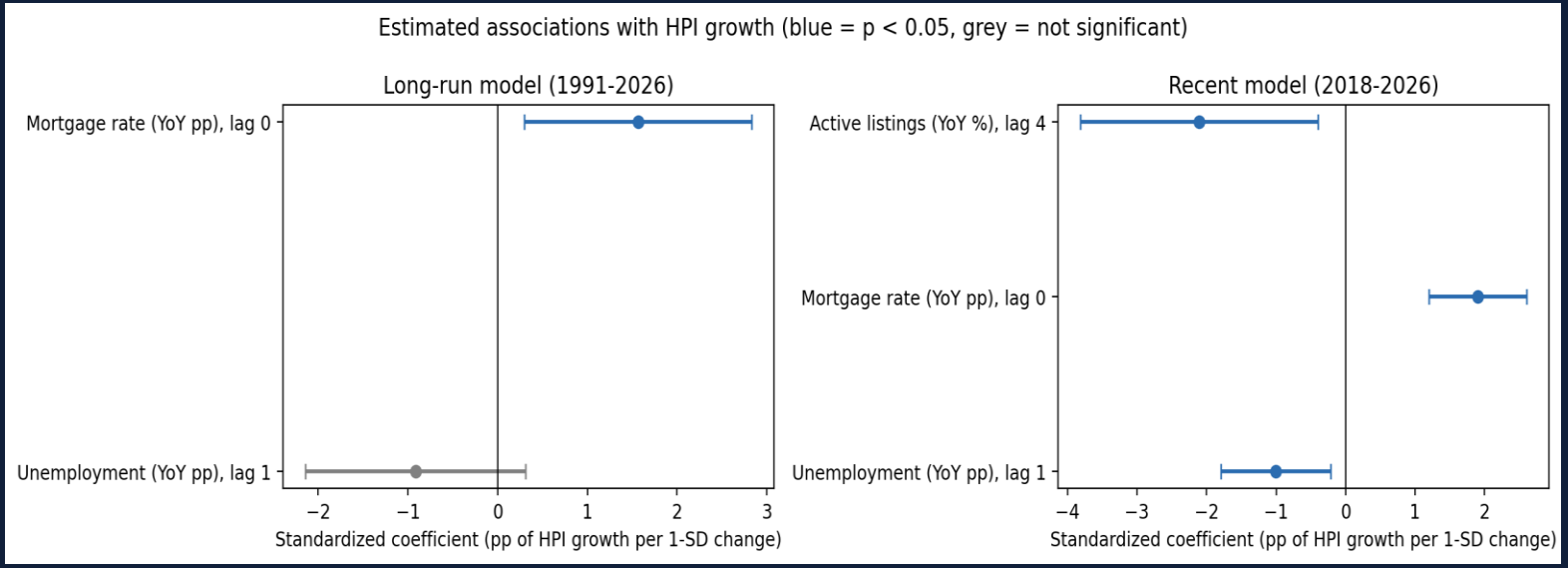
Results: inventory and mortgage rates stand out

Model	Predictor, lag	Estimate	95% interval	p
Recent	Listings, lag 4	−2.11	[−3.81, −0.40]	0.017
Recent	Mortgage, lag 0	+1.90	[+1.19, +2.60]	< 0.001
Recent	Unemployment, lag 1	−1.01	[−1.79, −0.22]	0.014
Long-run	Mortgage, lag 0	+1.56	[+0.30, +2.83]	0.016
Long-run	Unemployment, lag 1	−0.92	[−2.14, +0.31]	0.142

ANSWER TO OUR QUESTION

Mortgage rates are the sturdiest factor, moving with prices in the same quarter. Listings have the largest recent estimate, about a year earlier, but lose significance once the pandemic is controlled for.

Coefficient plot with 95% intervals



Robustness Checks:

Equal-strength tests: Evidence does not establish whether listings or mortgage rates have the stronger association (p = 0.850).

Newey-West Settings: Every significant result stays below 0.05. Conclusions don't change with different standard-error settings.

Next steps: test how well the findings travel

01 **Test on later quarters**

Check whether the model predicts quarters it has not seen, a first step toward forecasting.

02 **Match geography and fix the window**

Use metro unemployment and set the study window and lag range in advance.

03 **Add more predictors**

Try other local factors, such as new building permits, household income, etc.

NEW QUESTIONS

Why do mortgage rates and price growth move together? Is it the broader economy?

Is the delay for listings longer than a year?

Listings' correlation was getting stronger at lag 4, so the best delay might be longer.

Would the listings result hold with more post-pandemic data?

References and acknowledgements

[1] FHFA, “All-Transactions House Price Index for Charlottesville, VA (MSA),” FRED, ATNHPIUS16820Q. fred.stlouisfed.org/series/ATNHPIUS16820Q

[2] Freddie Mac, “30-Year Fixed Rate Mortgage Average in the United States,” FRED, MORTGAGE30US. fred.stlouisfed.org/series/MORTGAGE30US

[3] BLS, “Unemployment Rate in Charlottesville City, VA,” FRED, VACHAR0URN. fred.stlouisfed.org/series/VACHAR0URN

[4] Realtor.com, “Housing Inventory: Active Listing Count in Charlottesville, VA (CBSA),” FRED, ACTLISCOU16820. fred.stlouisfed.org/series/ACTLISCOU16820

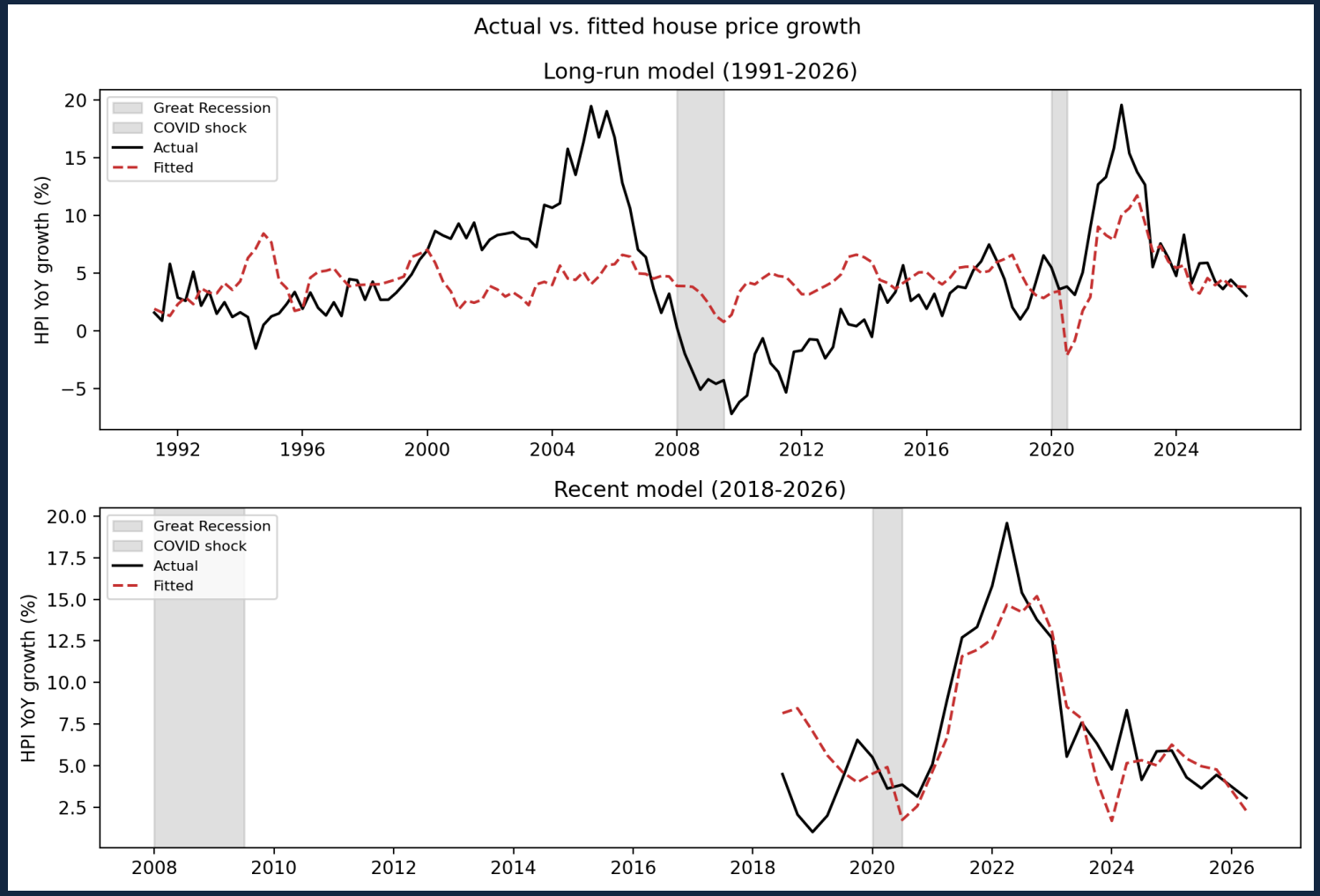
[5] W. K. Newey and K. D. West, Econometrica, vol. 55, no. 3, pp. 703–708, 1987. doi: 10.2307/1913610.

Repository: github.com/hershiaytoda/DS4002-Project2

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More inventory, slower growth?

Actual vs fitted annual house-price growth



Questions?